Coordinate Geometry Expertise

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10 questions 51 marks generated 2026-05-01 10:18

Question 1 **5 marks**

20 (a) Express $2x^2 - 11x + 9$ in the form $a(x - b)^2 - c$ where a, b and c are numbers to be found.

.....

(3)

The curve **C** has equation $y = 2(x - 3)^2 - 11(x - 3) + 9$

The point P is the minimum point on **C**

(b) Find the coordinates of P

(..... , ...)

(2)

(Total for Question 20 is 5 marks)

Question 2 **3 marks**

20 The straight line **L** has equation $y = 4x + 7$

The straight line **M** is perpendicular to **L** and passes through the point with coordinates (8, 1)

Find an equation for **M**
Give your answer in the form $y = mx + c$

.....

(Total for Question 20 is 3 marks)

Question 3 **6 marks**

23 OAB is a triangle.

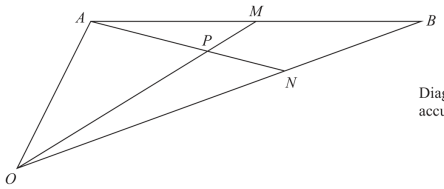

Diagram **NOT** accurately drawn

$\overrightarrow{OA} = 2\mathbf{a}$ and $\overrightarrow{OB} = 2\mathbf{b}$

M is the midpoint of AB.
N is the point on OB such that $ON : NB = 2 : 1$

P is the point on AN such that OPM is a straight line.

Use a vector method to find $OP : PM$
Show your working clearly.

.....

(Total for Question 23 is 6 marks)

Question 4 **7 marks**

22 *ABCD* is a kite, with diagonals *AC* and *BD*, drawn on a centimetre square grid, with a scale of 1 cm for 1 unit on each axis.

A is the point with coordinates (−3, 4)

The diagonals of the kite intersect at the point *M* with coordinates (0, 2)

Given that $AB = AD = 6.5$ cm and the x coordinate of *B* is positive,

find the coordinates of the points *B* and *D*.

(..... , ...)

(..... , ...)

(Total for Question 22 is 7 marks)

Question 5 **6 marks**

28 The curve **C** has equation $x^2 + y^2 = d - 11x$ where d is an integer.
The line **L** has equation $y = 3x + e$ where e is an integer.

C and **L** intersect at the point A and at the point B

The coordinates of A are (0.2, 2.6)

Work out the coordinates of B
Show your working clearly.

(..... , ...)

(Total for Question 28 is 6 marks)

Question 6 **5 marks**

22 The line with equation $y = x + 2$ intersects the curve with equation $x^2 + y^2 - 2y = 24$ at the points A and B.

Find the coordinates of A and B.
Show clear algebraic working.

(..... , ...)

(..... , ...)

(Total for Question 22 is 5 marks)

Question 7 **5 marks**

24 $\mathbf{L_1}$ and $\mathbf{L_2}$ are two straight lines.
The origin of the coordinate axes is O.

$\mathbf{L_1}$ has equation $5x + 10y = 8$
$\mathbf{L_2}$ is perpendicular to $\mathbf{L_1}$ and passes through the point with coordinates (8, 6)

$\mathbf{L_2}$ crosses the x-axis at the point A.
$\mathbf{L_2}$ intersects the straight line with equation $x = -3$ at the point B.

Find the area of triangle AOB.
Show your working clearly.

.....

(Total for Question 24 is 5 marks)

Question 8 5 marks

22 Here is a cuboid *ABCDEFGH*

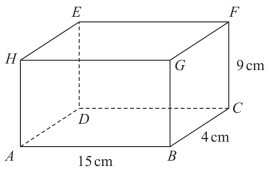

Diagram **NOT** accurately drawn

AB = 15 cm *BC* = 4 cm *CF* = 9 cm

(a) Work out the length of *BE*
Give your answer correct to 3 significant figures.

..... cm
(2)

Here is a cuboid *PQRSTUVW*

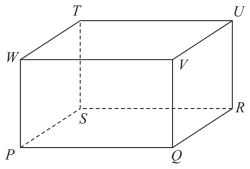

Diagram **NOT** accurately drawn

PR = 42 cm

The size of the angle between *PU* and the plane *PQRS* is 30°

M is the midpoint of *PR*

(b) Work out the size of angle *UMR*
Give your answer correct to 3 significant figures.

.....°
(3)

(Total for Question 22 is 5 marks)

Question 9 **4 marks**

22 *ABCD* is a rhombus.

The diagonals, *AC* and *BD*, intersect at the point *M*.
The coordinates of *M* are (6, −11)

The points *A* and *C* both lie on the line with equation $2y + 7x = 20$

Find the exact coordinates of the point where the line through *B* and *D* intersects the *y*-axis.

(..... , ...)

(Total for Question 22 is 4 marks)

Question 10 **5 marks**

22 ABC is a triangle in which angle $ABC = 90°$

p and q are integers such that

the coordinates of A are $(p, 10)$
the coordinates of B are $(-1, -5)$
the coordinates of C are $(8, q)$

Given that the gradient of AC is $-\frac{6}{7}$

work out the value of p and the value of q

$p =$ .....

$q =$ .....

(Total for Question 22 is 5 marks)